Liangbing Zhao

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Research Interest

- **Controllable Generation**: Developing generative models that enable fine-grained control over visual generation and editing through structured guidance, feedback-driven refinement, and physical priors.
- **World Models**: Building world models that learn physical dynamics through perception, interaction, and feedback, enabling prediction and planning in real-world environments.

Education

- **King Abdullah University of Science and Technology**, Saudi Arabia. Aug. 2024 – Dec. 2025
  M.Sc., Computer Science, supervised by Prof. Mohamed Elhoseiny.
  M.S./Ph.D Program. GPA: 3.71/4.0
  Thesis title: Toward Reflective and Self-Improving Generative Models
- **Beihang University**, China. Sept. 2019 – Jun. 2024
  B.S., Computer Science and Technology, supervised by Prof. Si Liu.

Experience

- **Visiting Research Student**, KAUST. Jul. 2023 - Mar. 2024, Saudi Arabia
  Prof. Mohamed Elhoseiny's group
  Structured image generation with diffusion bridge
- **Computer Vision Research Engineer**, Meitu. Jul. 2022 - May 2023, China
  Dr. Xuecheng Nie, MT Lab
  Video generation with hierarchical diffusion model

Publications

1. **Liangbing Zhao**, L. Zhuo, S. Paul, H. Li, and M. Elhoseiny, "From statics to dynamics: Physics-aware image editing with latent transition priors," **[ICML 2026]**.
2. S. Wang*, J. Zhao*, T. Pu*, **Liangbing Zhao***, X. Guo, Y. Cheng, C. Li, W. Ma, C. Tang, Z. Xu, et al., "Ultrafast visual perception beyond human capabilities enabled by motion analysis using synaptic transistors," **[Nature Communications 2026]**.
3. L. Zhuo*, **Liangbing Zhao***, S. Paul, Y. Liao, R. Zhang, Y. Xin, P. Gao, M. Elhoseiny, and H. Li, "From reflection to perfection: Scaling inference-time optimization for text-to-image diffusion models via reflection tuning," **[ICCV 2025]**.
4. Z. Yang, J. Chen, D. Xu, J. Fei, X. Shen, **Liangbing Zhao**, C.-M. Feng, and M. Elhoseiny, "Wikiautogen: Towards multi-modal wikipedia-style article generation," **[ICCV 2025]**.
5. E. M. Bakr, **Liangbing Zhao**, V. T. Hu, M. Cord, P. Perez, and M. Elhoseiny, "Toddlerdiffusion: Interactive structured image generation with cascaded schrödinger bridge," **[ICLR 2025]**.

Skills

- **Languages**: Chinese, English (CET-4, CET-6, TOEFL 103).
- **Coding**: C++, Python, JAVA, LATEX, Git, Pytorch

Awards

- KAUST Graduate Scholarship. 2024 - present
- Learning Merit Scholarship. 2022
- Beihang University Excellent Student. 2020
- Special Prize for Learning Merit Scholarship. 2020